Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

Treatment Patterns and Sequences of Pharmacotherapy for Patients Diagnosed with Depression in the United States: 2014 through 2019

David M. Kern (dkern2@its.jnj.com)
Janssen R&D https://orcid.org/0000-0001-5417-3925

M. Soledad Cepeda
Janssen Research and Development

Frank Defalco
Janssen Research and Development

Mila Etropolski
Janssen Research and Development

Research article

Keywords: Depression, treatment patterns, antidepressants, real-world evidence

Posted Date: October 15th, 2019

DOI: https://doi.org/10.21203/rs.2.16047/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Version of Record: A version of this preprint was published on January 3rd, 2020. See the published version at https://doi.org/10.1186/s12888-019-2418-7.

Abstract

Background : Understanding how patients are treated in the real-world is vital to identifying potential gaps in care. We describe the current pharmacologic treatment patterns for the treatment of depression.

Methods : Patients with depression were identified from four large national claims databases during 1/1/2014-1/31/2019. Patients had ≥2 diagnoses for depression or an inpatient hospitalization with a diagnosis of depression. Patients were required to have enrollment in the database ≥1 year prior to and three years following their first depression diagnosis. Treatment patterns were captured at the class level and included selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors, tricyclic antidepressants, other antidepressants, anxiolytics, hypnotics/sedatives, and antipsychotics . Treatment patterns were captured during all available follow-up.

Results : We identified 269,668 patients diagnosed with depression . The proportion not receiving any pharmacological treatment during follow-up ranged from 31% to 54%. Of the treated, approximately half received ≥2 different classes of therapy, a quarter received ≥3 classes and 10% received 4 or more. SSRIs were the most common first-line treatment; however, many patients received an anxiolytic, hypnotic/sedative, or antipsychotic prior to any antidepressive treatment. Treatment with a combination of classes was relatively uncommon across all treatment lines.

Conclusions : Many patients diagnosed with depression go untreated and many others receive a non-antidepressant medication class as their first treatment. More than half of patients received more than one type of treatment class during the study follow up, suggesting that the first treatment received may not be optimal for most patients.

Background

Major depressive disorder is highly prevalent in the United States, affecting more than 7% of adults and 13% of adolescents [1]. Treatment options for depression include psychotherapy, pharmacotherapy and invasive treatments such as electroconvulsive therapy (ECT) or transcranial magnetic stimulation (TMS) [2]. Treatment algorithms exist which aim to provide guidance to physicians in making effective treatment decisions to improve the chances of achieving a response [3, 4]; however, these algorithms do not provide granularity about specific medications to prescribe as there are many different treatment choices that fit within the algorithm structure, though starting with a selective serotonin reuptake inhibitor (SSRI) is the most commonly recommended approach. For patients treated with pharmacotherapy, only one quarter achieve full response and remission, while a similar proportion achieve no response, and the remainder achieve partial response or response without remission [2].

Currently, there is a lack of detail about how patients diagnosed with depression are treated in the real-world. Understanding treatment patterns in a real-world setting, as opposed to trials in which the patterns are predetermined [5], is an important step to improving and understanding gaps in care. For example,

atypical antipsychotics are the only approved adjunctive therapy for the treatment of depression [6, 7], but it is unknown how often these are used or how often other off-label adjunctive treatments are used.

This paper aims to fill the gaps in knowledge about real-world pharmacologic treatment patterns of patients diagnosed with depression and the role adjuvant therapy plays.

Methods

Depression cohort

We identified patients diagnosed with depression from large national insurance claims databases during 1/1/2014 through 1/31/2019. More detail about the databases is found in the 'Data Source' section below. Depression was defined as having two outpatient diagnoses within 365 days of each other or one inpatient diagnosis of depression according to the ICD–9-CM codes (269.2*, 269.3*, 300.4, 311) validated by Solberg et al [8] or the corresponding ICD–10-CM codes (F32.*, F33.*, F34.1, F53.0). No exclusion for comorbid conditions was applied.

The index date was the first observed medical claim in the database with a diagnosis of depression. Patients were required to have continuous enrollment in the database at least one year prior to and three years following the index date. Patients were excluded if they had evidence of treatment for depression—with an antidepressant or another treatment class of interest as defined below—more than 30 days prior to index.

Treatments and sequencing

Treatment patterns were captured at the class level and included SSRIs, serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), monoamine oxidase inhibitors (MAOIs), other antidepressants (including bupropion and trazodone, among others), anxiolytics, hypnotics/sedatives, and antipsychotics. Treatment sequences were captured during 30 days prior to the depression index date through all available follow-up, a minimum of 3 years. The term "treatment line" is used to describe the sequence of medication classes and combinations of medication classes received by patients during this time. Use of a specific medication class was captured at the first instance and not counted again in later lines of therapy—for example an individual filling an SSRI, switching to an SNRI, and then moving back to an SSRI would only be captured as switching from SSRI to SNRI. Because the analysis is at the class-level, in-class switching and in-class combination therapy is not captured. Drug eras were calculated as the time from the first fill for a drug in a medication class until discontinuation of that medication, allowing for gaps of up to 30 days beyond the days supply of a prescription (*Figure 1)*.. Combination therapy with multiple classes was defined as having at least 30 days of overlap in drug eras of more than one treatment class.

Data Source

The analysis was executed in four US-based administrative claims databases. Each database contains data from adjudicated health insurance claims (e.g., inpatient, outpatient/emergency department, and outpatient pharmacy) and health plan enrollment information. Briefly, the four databases included in this study were:

- IBM MarketScan® Commercial Database (CCAE): Includes data from 142 million individuals enrolled in employer-sponsored insurance health plans. At the time of this study data were available through January 31, 2019.
- IBM MarketScan® Multi-State Medicaid Database (MDCD): A claims database for 26 million Medicaid enrollees from multiple states. At the time of this study data were available through June 30, 2018.
- IBM MarketScan® Medicare Supplemental Database (MDCR): Includes data for more than 9 million retirees with primary or Medicare supplemental coverage through privately insured fee-for-service, point-of-service, or capitated health plans. At the time of this study data were available through January 31, 2019.
- Optum© De-Identified Clinformatics® Data Mart Database. Includes 84 million members with private health insurance, who are fully insured in commercial plans or in administrative services only and Medicare Advantage (Medicare Advantage Prescription Drug coverage. The population is representative of US commercial claims patients (0–65 years old) with some Medicare (65+ years old). At the time of this study data were available through December 31, 2018.

Data elements included were outpatient pharmacy dispensing claims (coded with National Drug Codes), inpatient and outpatient medical claims which provide diagnosis codes (coded in ICD–9-CM or ICD–10-CM) associated with a visit. The use of the IBM MarketScan and Optum claims databases was reviewed by the New England Institutional Review Board (IRB) and was determined to be exempt from broad IRB approval, as this research project did not involve human subjects research.

Common data model

Data from all four databases were mapped to standard concepts according to the Observational Medical Outcomes Partnership (OMOP) Common Data Model v5.0 [9] and the treatment sequence analysis was performed within the Observational Health Data Sciences and Informatics (or OHDSI, pronounced “Odyssey”) framework. Use of the common data model allowed for consistent and precise replication across each of the four databases.

Results

A total of 269,668 patients were included from across the four databases (*Table 1)..* The databases represent a wide range of ages, with the youngest patients from the CCAE and MDCD, while older individuals are more heavily represented in the MDCR and Optum databases. Females accounted for 62% of the patient population. Comorbid conditions such as pain, inflammatory disorders, hyperlipidemia, hypertension and diabetes were common in this population, particularly in the older Medicare population. A prior history of drug dependence was more common in the Medicaid patients (15.6%) compared with the other databases (6.5%–10.2%), and prior evidence of suicidal thoughts was found in more than 1 in 25 Medicaid patients versus the lowest prevalence of 1 in 100 individuals from Medicare.

Roughly one-third of patients from the CCAE (30.6%), MDCR (33.6%), and Optum (36.6%) databases did not receive any antidepressant or related medication during the entire follow-up period, while more than half of patients in the MDCD data (53.6%) were untreated with pharmacotherapy (*Table 2)..* Of patients who did receive a treatment, approximately half went on to receive a second treatment class (range across databases: 46.5%–57.5%), a quarter received a third (23.4%–29.0%), and one in ten received a fourth (9.6%–13.5%).

The most common medication class used during first-line therapy was SSRI, however there was variability in their use—half of patients from the CCAE database received SSRI as monotherapy or part of a combination first line treatment (50.0%) compared with just one-third of patients in the MDCD database (33.1%) (*Table 3)..* Non-antidepressant use made up a significant share of first-line treatments, with anxiolytics as the next most common first line treatment class received in all databases except for MDCD where hypnotic/sedative use was found in 23.3% of patients. Use of antipsychotics was not an uncommon first line approach, nearly 12% of Medicaid patients and more than 5% of all other patients received an antipsychotic as their initial treatment.

Combination therapy with two distinct medication classes was used as the first line of treatment for 6.4%–11.2% of patients, and the prevalence of combination therapy did not markedly increase in later lines of therapy as might be expected when treatments fail to be effective (*Table 4)..* Patients from the MDCD data had the highest rates of combination therapy while those in MDCR were least likely to receive medications from two distinct classes simultaneously.

Within patients receiving monotherapy SSRI as their initial therapy, more than half of patients went on to receive a second treatment class (*Table 5)..* Combination of an SSRI plus another treatment class was found in 3.2%–4.8% of patients receiving second line therapy; however, use in combination with an antipsychotic—the only approved treatment class for adjuvant depression treatment—was found in a minority of patients (0.6%–1.2%).

The sequence of treatments within each database are shown in *Figure 2.* This figure illustrates that while SSRI use is the most common first line treatment, the use of non-antidepressants—particularly anxiolytics, hypnotics/sedatives, and anticonvulsants—are common. Approximately half of patients starting on an SSRI never fill another class, while the other half moves on to a variety of different therapies and no single specific class appears to be the dominant second choice. The proportion of

treatments accounted for by SSRIs decreases at each successive treatment line, while anticonvulsant, antipsychotic, and SNRI use steadily increases (*Figure 3)*,; a pattern that is relatively consistent in each of the databases.

Discussions

There has been limited prior research examining depression treatment sequences and the current work substantially expands upon this groundwork. Gauthier et al [10] examined patterns of switches, combinations, dose escalation, and discontinuation of antidepressants in general, but did not look at individual drugs or classes, and did not include non-antidepressant treatment classes. Hripcsak et al. [11], used a similar methodology to identify depression treatment sequences in multiple databases; however, this analysis did not capture combination therapy or use of non-antidepressant medication classes, among other differences in the approach.

Our study leveraged prescription claims data from four patient populations representing a broad cross-section of the US population, including commercially insured individuals, those receiving Medicare, and individuals on Medicaid. There was a relatively low prevalence of first-line SSRI use (occurring in less than half of patients) in contrast to many of the treatment guidelines recommending starting with monotherapy SSRI [12, 13]. Furthermore, use of anxiolytics, anticonvulsants, and hypnotics/sedatives were commonly used as the first treatment choice in these patients newly diagnosed with depression, potentially pointing to a reluctancy of physicians to prescribe antidepressants [14] but being more comfortable using other classes of drugs such as anxiolytics [15, 16]. High use of benzodiazepines, which comprised the majority of anxiolytic use in this study, is concerning because they are not recommended as a first-line therapy [12] and they carry concerns of abuse [17–19] and risk of overdose [20, 21].

This study showed that while general trends across these populations were relatively similar there were some important differences. Specifically, patients covered by Medicaid tended to have treatment patterns that were different than the other three groups—more than half of patients diagnosed with depression were untreated, first-line SSRI use was lower, use of two distinct treatment classes in combination was more common, and use of alternative treatment classes outside of antidepressants occurred more often. The Medicaid sample represents a population of vulnerable individuals of lower socio-economic status and high burden of disease and it appears they are receiving different care when compared with the other patient populations.

Combination therapy was a relatively uncommon practice, occurring in approximately 10% of individuals consistently across each of the first four treatment lines. It was expected that combination therapy would increase in later lines of therapy as treatment guidelines recommend to augment a current therapy with an additional medication if a patient does not achieve full response [12].

The high prevalence of non-antidepressant treatment classes could reflect the high rates of comorbid conditions, such as anxiety disorder or sleep disorders [22]; however, these medications are largely being

prescribed as monotherapy and not in combination with an antidepressant.

Many patients received no pharmacotherapy for their depression during the entire follow-up, a period covering a minimum of three years in all patients. This was not limited to only Medicaid patients, as mentioned above, but also affected approximately a third of patients from the other databases. This is likely a vast underestimate of the true prevalence of untreated depression patients, because a significant proportion of individuals go undiagnosed and therefore are not able to receive treatment. Previous research screening individuals for depression rather than relying on a physician diagnosis has found that just 29%–46% received a treatment for their depression [23, 24].

Limitations

There are limitations to this study. This analysis focused only on pharmacotherapy for the treatment of depression and did not examine rates of psychotherapy or procedures such as ECT or TMS, which play an important role in the overall care patients receive, and may account for the proportion of patients that were classified as untreated. Patients with depression were identified using diagnoses codes which are not a perfect tool; however, we used a previously published algorithm for identifying depression in claims data which achieved high validity (PPV = 99%) [8]. This analysis did not capture any within-class switching or combination; for example, receiving two SSRIs is simply captured as monotherapy SSRI use and switching from one SSRI to another does not appear as a change in therapy. There is no diagnosis associated with prescription claims, thus receiving treatment for non-antidepressant classes is not guaranteed to have been prescribed for treating the underlying depression or its symptoms. To mitigate this misclassification due to receiving therapy for reasons unrelated to depression, we required treatment to occur at the time of or following the first diagnosis of depression with no prior history of treatment.

Strengths

This study included more than a quarter-million individuals diagnosed with depression across four major claims databases representing a full-spectrum of ages and types of insurance coverage. When examined together, these databases provide generalizability to a broad cross-section of the United States. We were able to leverage the infrastructure of the common data model and the tools from the OHDSI network to achieve a uniform and consistent approach across each of these four databases whose underlying data structures differ. This work expands upon previous work by not limiting the analysis to only drugs that are specifically classified as antidepressants. It is widely known that medications in various other classes are commonly used to treat patients with depression and this study reflects real-world prescribing practices.

Conclusions

This study provides the most detailed reflection of real-world pharmacotherapy practices for the treatment of depression in the United States to date. Understanding how depression patients are being

managed is an important first step in improving care.

Declarations

Ethics approval and consent to participate

The use of the databases was reviewed by the New England Institutional Review Board (IRB) and was determined to be exempt from broad IRB approval, as this research project did not involve human subjects research.

Consent for publication

Not applicable

Availability of data and materials

Data are available on request

Competing interests

All authors are employees of Janssen Research & Development, LLC, which markets and develops treatments for depression, and are stockholders of Johnson & Johnson, Janssen's parent company.

Funding

Janssen Research & Development, LLC funded this research

Authors' contributions

DMK contributed to the study design, data acquisition and analysis, interpretation of results, and writing and revising the manuscript. MSC contributed to the study design, interpretation of results, and reviewing the manuscript. FD contributed to data acquisition, data analysis, and reviewing of the manuscript. ME contributed to the study design, interpretation of results, and reviewing the manuscript.

Acknowledgements

Not applicable

Abbreviations

CCAE: IBM MarketScan® Commercial Database

ECT: Electroconvulsive therapy

ICD–10-CM: International Classification of Diseases, Tenth Revision, Clinical Modification

ICD–9-CM: International Classification of Diseases, Ninth Revision, Clinical Modification

IRB: Institutional Review Board

MAOI: Monoamine oxidase inhibitor

MDCD: IBM MarketScan® Multi-State Medicaid Database

MDCR: IBM MarketScan® Medicare Supplemental Database

OHDSI: Observational Health Data Sciences and Informatics

OMOP: Observational Medical Outcomes Partnership

SNRI: Serotonin and norepinephrine reuptake inhibitor

SSRI: Selective serotonin reuptake inhibitor

TCA: Tricyclic antidepressant

TMS: Transcranial magnetic stimulation

Tables

e 1. Characteristics of patients from each of the four study databases

aracteristic	CCAE (N=114,543)	MDCD (N=69,006)	MDCR (N=5,660)	Optum (N=80,459)
e at index (years), Mean (SD)	32.4 (15.6)	37.9 (25.7)	74.5 (8.5)	51.0 (23.5)
nder: Female, %	60.4%	65.6%	60.5%	62.4%
arlson comorbidity index score *, Mean (SD)	0.5 (1.3)	1.9 (2.9)	2.9 (2.6)	1.7 (2.5)
*nmon comorbid conditions**				
Pain	46.6%	58.7%	68.8%	59.1%
nflammatory disorder of the respiratory system	33.8%	38.2%	31.4%	33.4%
anxiety disorder	25.7%	21.3%	19.8%	27.3%
Traumatic injury	21.6%	27.4%	30.8%	25.4%
arthropathy	20.3%	29.8%	51.9%	35.9%
Dysthymia	20.3%	8.6%	17.4%	15.7%
nflammatory disorder of digestive system	20.2%	24.1%	18.0%	19.1%
Neoplastic disease	14.8%	10.2%	43.6%	24.3%
Hyperlipidemia	13.2%	23.7%	58.0%	38.0%
Visual system disorder	12.7%	31.3%	52.8%	31.7%
Hypertensive disorder	12.7%	36.6%	67.6%	39.2%
Malaise and fatigue	12.3%	11.9%	22.5%	19.2%
Backache	10.8%	16.9%	20.1%	17.8%
Vascular disorder	8.1%	19.9%	45.9%	24.8%
Drug dependence	6.5%	15.6%	8.0%	10.2%
Osteoarthritis	5.5%	16.5%	34.1%	19.6%
Diabetes mellitus	5.0%	18.7%	26.7%	15.7%
cidal history				
Suicidal thoughts	3.9%	4.3%	1.0%	2.9%
Suicidal deliberate poisoning	0.2%	0.2%	0.0%	0.2%

* The Charlson Comorbidity Index and individual comorbidities were captured during the 365 days preceding and including the index date.

Abbreviations: CCAE, IBM MarketScan® Commercial Database; MDCD, IBM MarketScan® Multi-State Medicaid Database; MDCR, IBM MarketScan® Medicare Supplemental Database; SD, standard deviation

Table 2. Proportion of patients who were untreated or received, at least one, two, three, or four distinct treatment lines during the entire follow-up period

	CCAE (N=114,543)		MDCD (N=69,006)		MDCR (N=5,66)		Optum (N=80,459)	
Treatment line	% of all patients	% of treated	% of all patients	% of treated	% of all patients	% of treated	% of all patients	% of treated
Untreated	30.6%		53.6%		33.6%		36.6%	
1st	69.4%	100%	46.4%	100%	66.4%	100%	63.4%	100%
2nd	39.9%	57%	21.6%	46%	35.1%	53%	33.4%	53%
3rd	20.2%	29%	10.9%	23%	16.1%	24%	16.2%	26%
4th	9.4%	14%	5.0%	11%	6.4%	10%	7.2%	11%

Abbreviations: CCAE, IBM MarketScan® Commercial Database; MDCD, IBM MarketScan® Multi-State Medicaid Database; MDCR, IBM MarketScan® Medicare Supplemental Database

Table 3. Proportion of patients treated with each medication class out of those receiving first-line therapy (includes mono- or combination therapy)

	CCAE (N=79,490)		MDCD (N=32,046)		MDCR (N=3,758)		Optum (N=51,023)	
SSRI	39,707	50.0%	10,620	33.1%	1,438	38.3%	22,183	43.5%
Anxiolytic	14,003	17.6%	5,935	18.5%	652	17.3%	8,511	16.7%
Other antidepressant	11,098	14.0%	3,235	10.1%	444	11.8%	6,497	12.7%
Hypnotic/Sedative	6,789	8.5%	7,468	23.3%	473	12.6%	6,617	13.0%
Anticonvulsant	7,310	9.2%	3,907	12.2%	513	13.7%	5,641	11.1%
SNRI	3,860	4.9%	803	2.5%	218	5.8%	2,514	4.9%
Antipsychotic	4,242	5.3%	3,771	11.8%	250	6.7%	2,642	5.2%
Tricyclic	1,161	1.5%	720	2.2%	53	1.4%	684	1.3%
MAOI	6	0.0%	2	0.0%	4	0.1%	16	0.0%

Abbreviations: CCAE, IBM MarketScan® Commercial Database; MDCD, IBM MarketScan® Multi-State Medicaid Database; MDCR, IBM MarketScan® Medicare Supplemental Database; SSRI, selective serotonin reuptake inhibitor; SNRI, serotonin and norepinephrine reuptake inhibitor; MAOI, monoamine oxidase inhibitor

Table 4. Prevalence of combination therapy during each treatment line within treated patients

Treatment line	CCAE	MDCD	MDCR	Optum
1st	9.1%	11.2%	6.4%	7.2%
2nd	10.2%	12.0%	7.8%	10.0%
3rd	10.4%	10.7%	9.0%	10.1%
4th	10.5%	11.2%	6.9%	10.6%

Abbreviations: CCAE, IBM MarketScan® Commercial Database; MDCD, IBM MarketScan® Multi-State Medicaid Database; MDCR, IBM MarketScan® Medicare Supplemental Database

Table 5. Treatment patterns of those receiving first line monotherapy SSRI treatment

	CCAE		MDCD		MDCR		Optum	
Received first line SSRI (monotherapy)	35,384		8,574		1,332		20,117	
Patients on first line SSRI monotherapy who went on to receive second treatment class	19,035	53.8%	4,727	55.1%	691	51.9%	10,138	50.4%
Second line is combination therapy	1,981	10.4%	474	10.0%	55	8.0%	965	9.5%
Second line is combination therapy which includes an SSRI	805	4.2%	149	3.2%	27	3.9%	483	4.8%
Second line is combination therapy of SSRI + antipsychotic	154	0.8%	59	1.2%	4	0.6%	79	0.8%

Abbreviations: CCAE, IBM MarketScan® Commercial Database; MDCD, IBM MarketScan® Multi-State Medicaid Database; MDCR, IBM MarketScan® Medicare Supplemental Database; SSRI, selective serotonin reuptake inhibitor

References

1. National Institute of Mental Health. Major Depression 2019 [Last Accessed: September 9, 2019]. Available from: https://www.nimh.nih.gov/health/statistics/major-depression.shtml. 2. Fava M, Rush AJ, Trivedi MH, Nierenberg AA, Thase ME, Sackeim HA, et al. Background and rationale for the sequenced treatment alternatives to relieve depression (STAR*D) study. The Psychiatric clinics of North America. 2003;26(2):457-94, x. 3. Adli M, Wiethoff K, Baghai TC, Fisher R, Seemuller F, Laakmann G, et al. How Effective Is Algorithm-Guided Treatment for Depressed Inpatients? Results from the Randomized Controlled Multicenter German Algorithm Project 3 Trial. The international journal of neuropsychopharmacology. 2017;20(9):721-30. 4. Trivedi MH, Kleiber BA. Algorithm for the treatment of chronic depression. The Journal of clinical psychiatry. 2001;62 Suppl 6:22-9. 5. Gaynes BN, Rush AJ, Trivedi MH, Wisniewski SR, Spencer D, Fava M. The STAR*D study: treating depression in the real world. Cleveland Clinic journal of medicine. 2008;75(1):57-66. 6. Wang P, Si T. Use of antipsychotics in the treatment of depressive disorders. Shanghai archives of psychiatry. 2013;25(3):134-40. 7. Wang S-M, Han C, Lee S-J, Jun T-Y, Patkar AA, Masand PS, et al. Second Generation Antipsychotics in the Treatment of Major Depressive Disorder: An Update. Chonnam Med J. 2016;52(3):159-72. 8. Solberg LI, Engebretson KI, Sperl-Hillen JM, Hroscikoski MC, O'Connor PJ. Are claims data accurate enough to identify patients for performance measures or quality improvement? The case of diabetes, heart disease, and depression. American journal of medical quality : the official journal of the American College of Medical Quality. 2006;21(4):238-45. 9. Stang PE, Ryan PB, Racoosin JA, Overhage JM, Hartzema AG, Reich C, et al. Advancing the science for active surveillance: rationale and design for the Observational Medical

Outcomes Partnership. Annals of internal medicine. 2010;153(9):600-6. 10. Gauthier G, Guérin A, Zhdanava M, Jacobson W, Nomikos G, Merikle E, et al. Treatment patterns, healthcare resource utilization, and costs following first-line antidepressant treatment in major depressive disorder: a retrospective US claims database analysis. BMC Psychiatry. 2017;17(1):222. 11. Hripcsak G, Ryan PB, Duke JD, Shah NH, Park RW, Huser V, et al. Characterizing treatment pathways at scale using the OHDSI network. Proceedings of the National Academy of Sciences of the United States of America. 2016;113(27):7329-36. 12. American Psychiatric Association. Practice Guidelines for the Treatment of Patients with Major Depressive Disorder 2010 [Last Accessed: August 19, 2019]. Available from: https://psychiatryonline.org/pb/assets/raw/sitewide/practice_guidelines/guidelines/mdd.pdf. 13. Gelenberg AJ. A review of the current guidelines for depression treatment. The Journal of clinical psychiatry. 2010;71(7):e15. 14. Hyde J, Calnan M, Prior L, Lewis G, Kessler D, Sharp D. A qualitative study exploring how GPs decide to prescribe antidepressants. The British journal of general practice : the journal of the Royal College of General Practitioners. 2005;55(519):755-62. 15. Haw C, Stubbs J. Benzodiazepines–a necessary evil? A survey of prescribing at a specialist UK psychiatric hospital. Journal of psychopharmacology (Oxford, England). 2007;21(6):645-9. 16. Sirdifield C, Anthierens S, Creupelandt H, Chipchase SY, Christiaens T, Siriwardena AN. General practitioners' experiences and perceptions of benzodiazepine prescribing: systematic review and meta-synthesis. BMC Family Practice. 2013;14(1):191. 17. Brett J, Murnion B. Management of benzodiazepine misuse and dependence. Aust Prescr. 2015;38(5):152-5. 18. Lader M. Benzodiazepines revisited–will we ever learn? Addiction (Abingdon, England). 2011;106(12):2086-109. 19. Votaw VR, Geyer R, Rieselbach MM, McHugh RK. The epidemiology of benzodiazepine misuse: A systematic review. Drug and alcohol dependence. 2019;200:95-114. 20. Park TW, Saitz R, Ganoczy D, Ilgen MA, Bohnert AS. Benzodiazepine prescribing patterns and drug overdose mortality among individuals receiving opioid analgesics. Addiction Science & Clinical Practice. 2015;10(1):A48. 21. Sun EC, Dixit A, Humphreys K, Darnall BD, Baker LC, Mackey S. Association between concurrent use of prescription opioids and benzodiazepines and overdose: retrospective analysis. BMJ. 2017;356:j760. 22. Cepeda MS, Stang P, Blacketer C, Kent JM, Wittenberg GM. Clinical Relevance of Sleep Duration: Results from a Cross-Sectional Analysis Using NHANES. Journal of clinical sleep medicine : JCSM : official publication of the American Academy of Sleep Medicine. 2016;12(6):813-9. 23. Olfson M, Blanco C, Marcus SC. Treatment of Adult Depression in the United States. JAMA Internal Medicine. 2016;176(10):1482-91. 24. Cepeda MS, Stang P, Makadia R. Depression Is Associated With High Levels of C-Reactive Protein and Low Levels of Fractional Exhaled Nitric Oxide: Results From the 2007-2012 National Health and Nutrition Examination Surveys. The Journal of clinical psychiatry. 2016;77(12):1666-71.

Figures

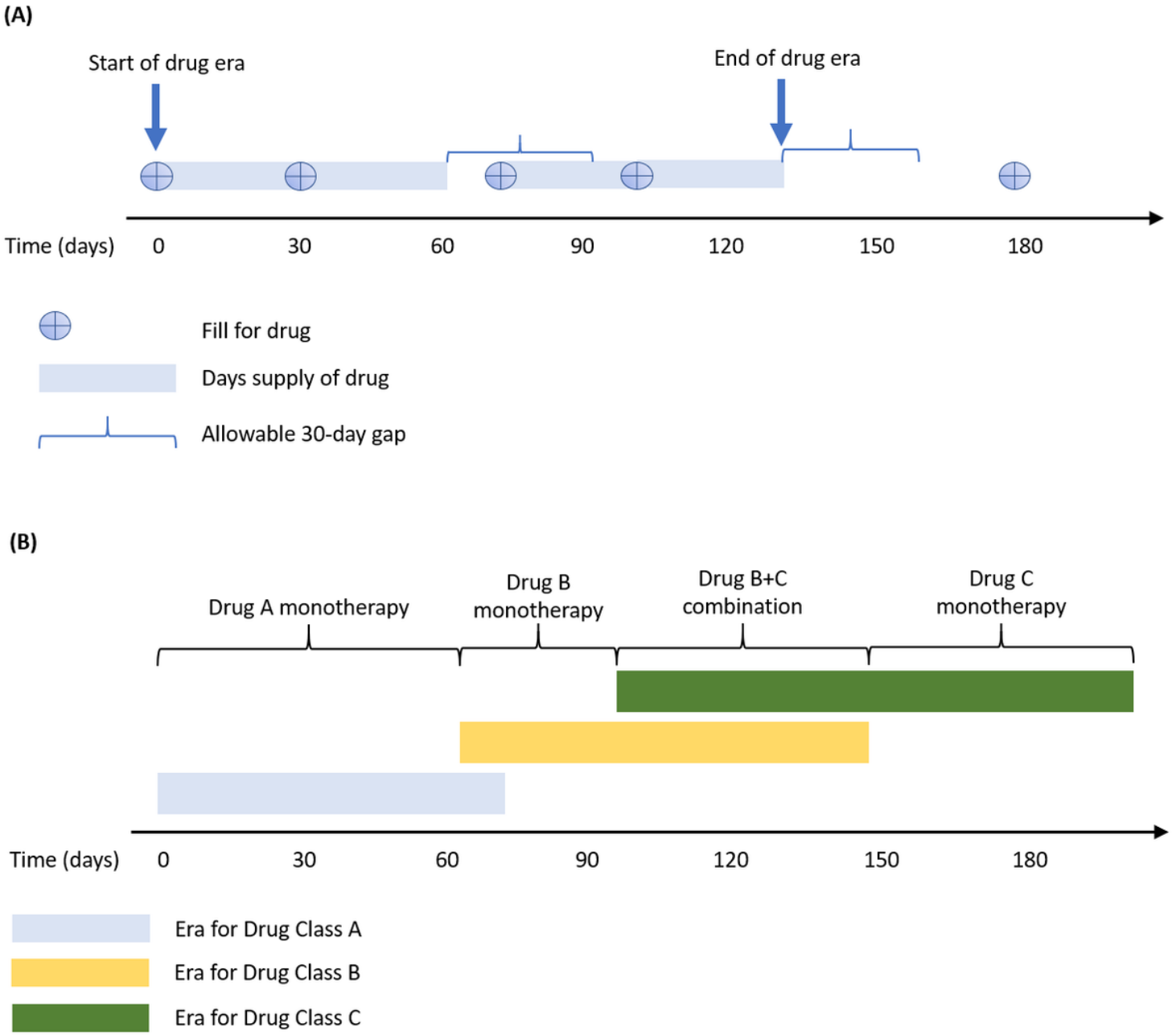

Figure 1

Study design illustration for drug eras, switching and combination therapy classification

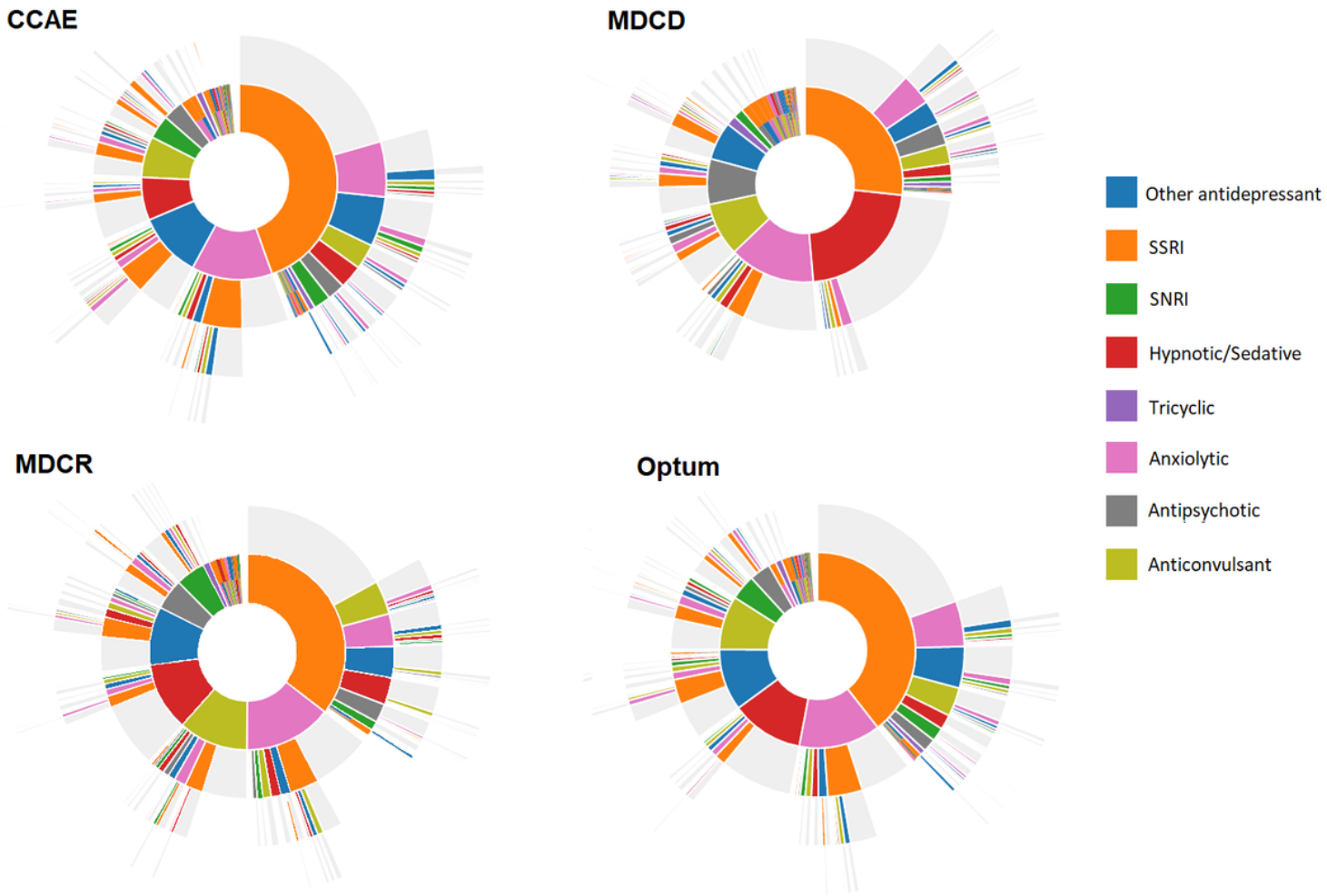

Figure 2

Sunburst of treatment patterns starting with first line (inner-most donut) to fourth line (outer slices)

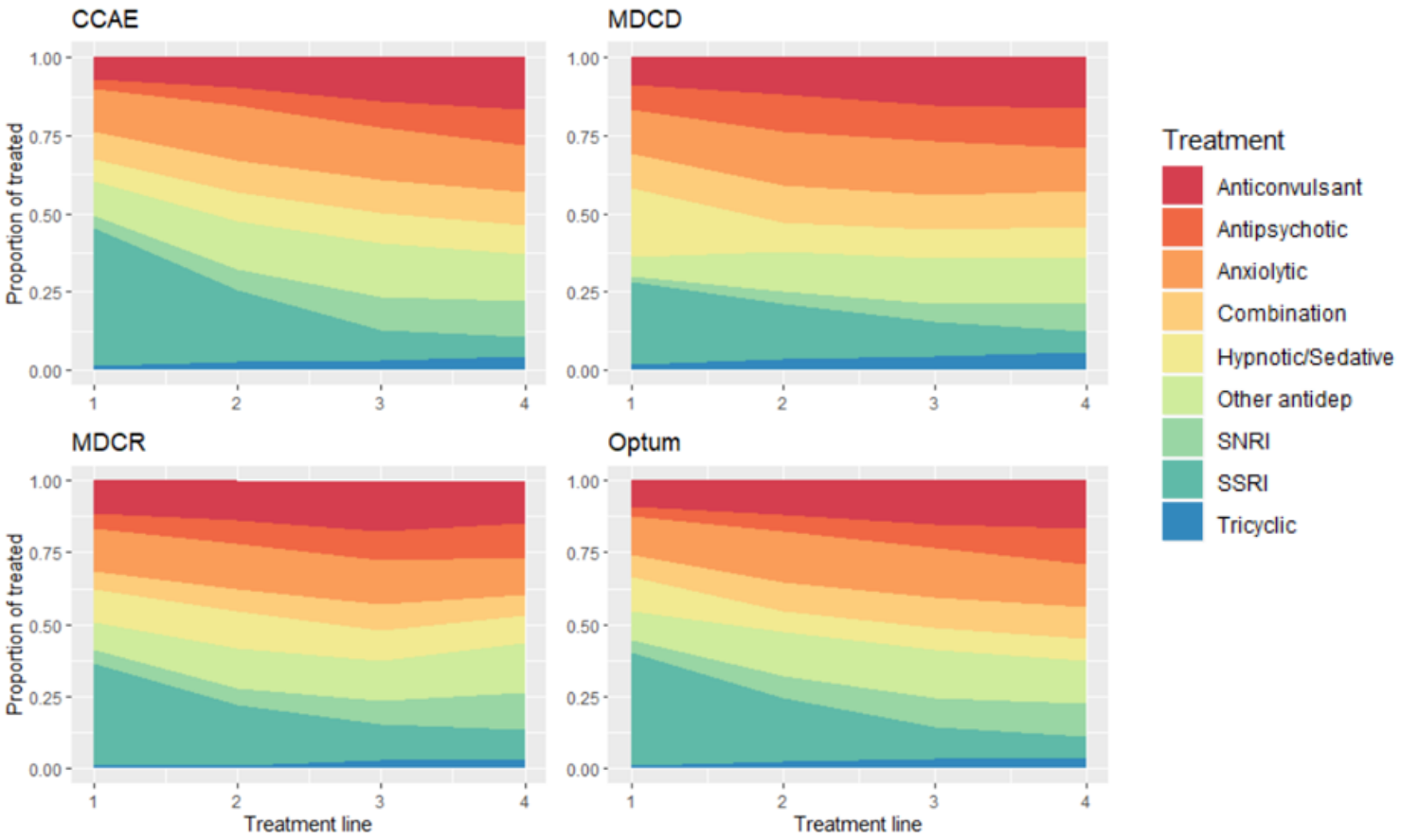

Figure 3

Proportional share of each treatment line by medication class